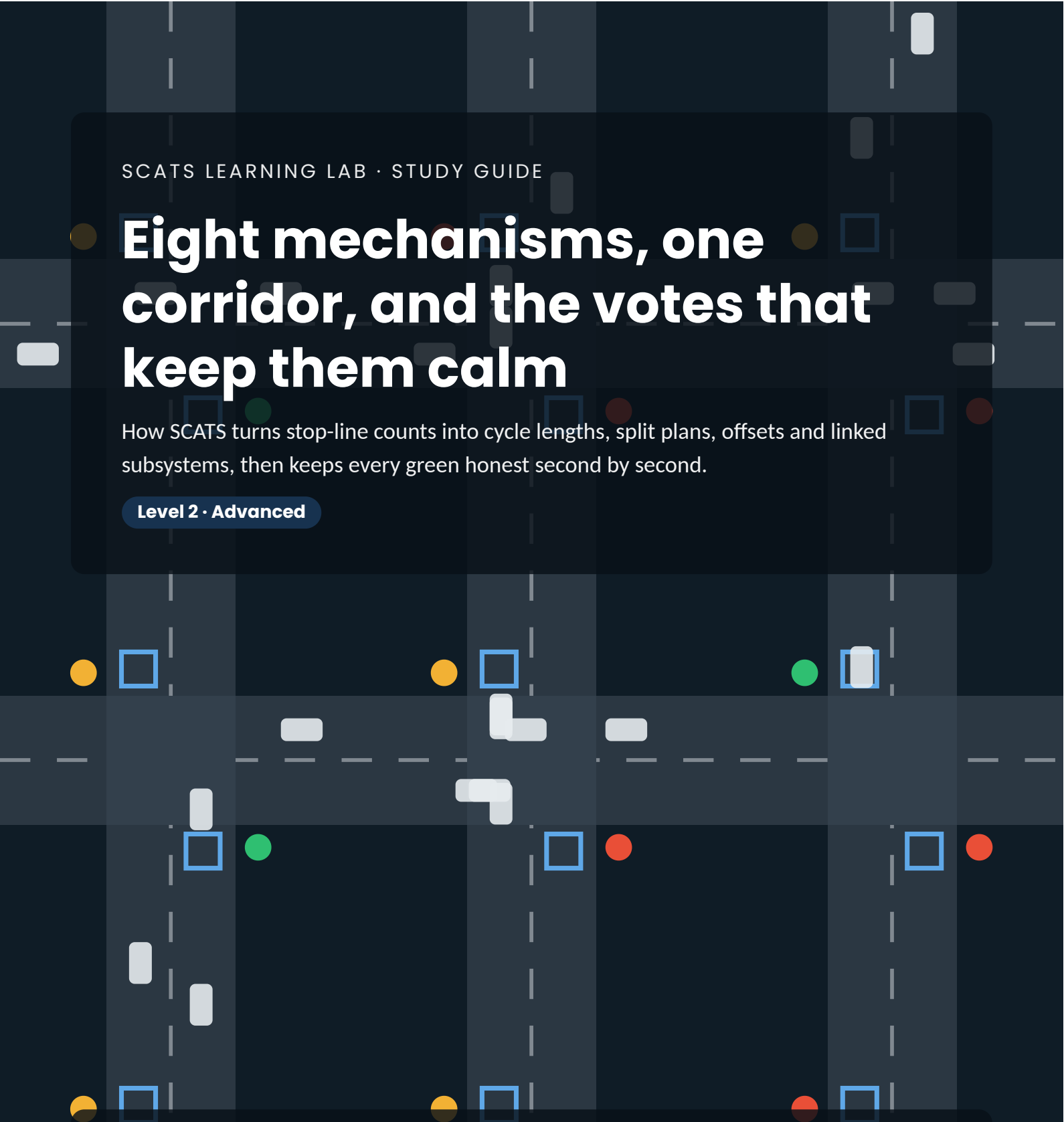
SCATS LEARNING LAB · STUDY GUIDE

Eight mechanisms, one corridor, and the votes that keep them calm

How SCATS turns stop-line counts into cycle lengths, split plans, offsets and linked subsystems, then keeps every green honest second by second.

Level 2 · Advanced



INSIDE

- Why one 'critical' junction sets the rhythm for its neighbours
- Cycle length by small steps: the arithmetic of C × DS ÷ 0.9
- Plan voting, marriage and divorce: decisions that refuse to flip-flop
- What happens when you switch each mechanism off, measured in SUMO

Beginner-friendly companion to smrahmanbd.github.io/scats-learning-lab

The map: SCATS Advanced

This guide assumes you have read the Foundation level, or at least know what a cycle, a phase, a split and an offset are, and that SCATS measures each green with the **degree of saturation (DS)**. Everything else is built from scratch with small numbers. There are eight ideas: the first five are **strategic** (decided once per cycle for groups of junctions), the last three are about **tactical** control, priority and the system as a whole.

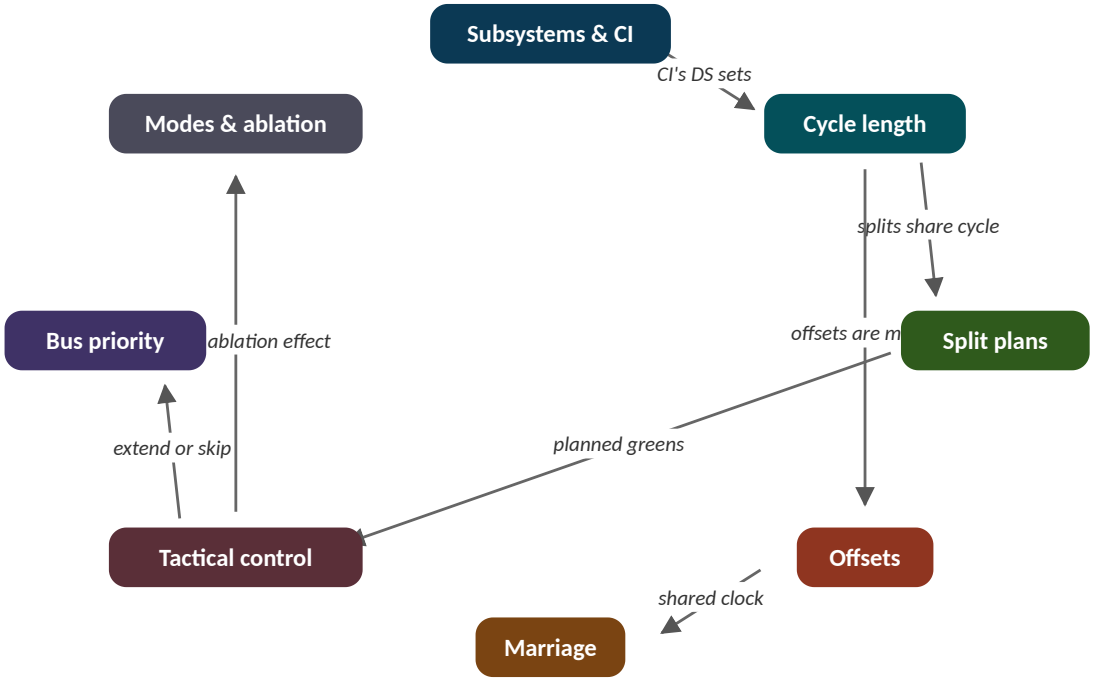


Figure 0. The whole guide on one page. Each box is one concept section; the arrows say how one idea feeds the next. Come back to this page whenever you feel lost.

1. Subsystems and the critical intersection
2. Cycle length: bounded steps towards a target DS
3. Split plans: equalise DS, then vote
4. Offsets and green waves
5. Marriage and divorce
6. Tactical control: gap-out, skipping, holding
7. Public transport priority
8. Operating modes and the full system on SUMO

HOW TO USE THIS GUIDE

Read one concept per sitting (20 to 30 minutes). Do every **Simple calculation** with a pencil before reading the answer. Say each **Picture this** scene out loud. The **Recap so far** box grows as you go; reading it is free revision. Finish with the self-quiz and the revision plan.

What you will be able to do

- Find the critical intersection of a subsystem from its DS values and explain why it sets the cycle.
- Compute the next cycle length from C, DS and the target, using the step limits and LMIN/LMAX.
- Predict each phase's DS under a different split plan and pick the plan that balances them.
- Work out offsets for an eastbound or westbound green wave and explain offset voting.
- Trace marriage and divorce of two subsystems through a run of link volumes.
- Decide when a green gaps out, when a phase is skipped, and why the main road is held.
- Compare high, medium and low transit priority for a bus that just missed its green.
- Name the operating modes and read the SUMO ablation: which mechanism matters most.

Before we start

HOOK

SCATS refuses to obey a single good result. Even when a new split plan is clearly better this cycle, SCATS waits until it wins 2 of the last 3 cycles. For offsets it waits for 4 of 5. Patience is designed in, because a plan that flips every cycle disturbs drivers more than it helps them.

HOOK

Junctions get married. Two neighbouring groups of signals can 'marry' to share one cycle and one clock, and 'divorce' when the traffic between them fades. The thresholds for marrying and divorcing are deliberately different, so they do not argue all day.

HOOK

Switching off cycle adaptation made our corridor slightly better. In the SUMO ablation, freezing the cycle at 80 s gave 34.4 s of delay instead of 36.9 s, because 80 s happened to suit that demand. Switching off tactical control, by contrast, raised delay to 62.0 s. Not every mechanism matters equally on every road.

Words you will meet

You do not need to memorise these now. Each one is explained again when it first matters.

Word	In plain words
Subsystem	A group of one to ten signals that share one cycle length so they can be coordinated.
Critical intersection (CI)	The busiest junction in a subsystem. Its DS decides the subsystem's cycle.
DS	Degree of saturation: how much of a green was really used by traffic. 0.9 means 90%.
VK	Reconstituted flow: SCATS's estimate of how many vehicles a green could have carried, from DS.
Split plan	A stored way of sharing the cycle between phases, for example 50% / 20% / 30%.
Offset plan	A stored pattern of start times along a road, for example 'green wave eastbound'.
Strategic control	Decisions made once per cycle by the regional computer: cycle, splits, offsets, linking.
Tactical control	Decisions made every second by the local controller: end a green early, skip a phase.
Hysteresis	Using a higher threshold to switch on than to switch off, so a decision does not flicker.

CONCEPT 1 OF 8 · SCATS ADVANCED

Subsystems and the critical intersection

HOOK

A city may have thousands of signals. SCATS does not try to give them one cycle; it splits them into small teams. Each team has a captain. Who should the captain be?



IN PLAIN WORDS

A rowing crew rows to one rhythm, and the stroke rower at the back sets it. If the stroke tires, the whole boat slows. The other rowers keep the same rhythm but each pulls with their own strength. *Where the picture breaks:* in SCATS the 'stroke' is not fixed: the CI is chosen by the engineer as the junction that usually needs the most time.

SCATS groups signals into **subsystems**, usually one to ten junctions that must share a cycle length to be coordinated. Each subsystem has one **critical intersection (CI)**, normally the busiest. The CI's DS decides the subsystem's cycle length.

The other junctions follow that cycle, but each has its own **split plan** (how it shares the cycle among its phases) and an **offset** (when its cycle starts compared with the CI).

Once per cycle the regional computer runs the same **strategic loop**: collect DS and VK from every strategic loop detector; move the cycle length; vote on split plans; vote on offset plans; decide whether neighbouring subsystems should marry or divorce.

$$\mathrm{DS} = (\mathrm{g} - (\mathrm{T} - \mathrm{n} \times \mathrm{h})) \div \mathrm{g}$$

g = green, T = unoccupied loop time, n = vehicles, h = standard space time (1.6 s here)

SIMPLE CALCULATION · Which junction is critical?

Subsystem J1-J2-J3. At J2 the main phase had 30 s of green, the loop was empty for 22 s in total, 12 vehicles crossed, h = 1.6 s. J1 has DS 0.78 and J3 has DS 0.66.

1. Space a busy stream needs: n × h = 12 × 1.6 = **19.2 s**.
2. Wasted green: T − n × h = 22 − 19.2 = **2.8 s**.
3. Used green: 30 − 2.8 = **27.2 s**.
4. DS at J2 = 27.2 ÷ 30 = **0.91**.
5. Compare: J1 0.78, J2 0.91, J3 0.66. The highest is **J2**.

J2 is the busiest and is the natural CI. Its 0.91 will drive the subsystem's next cycle length.

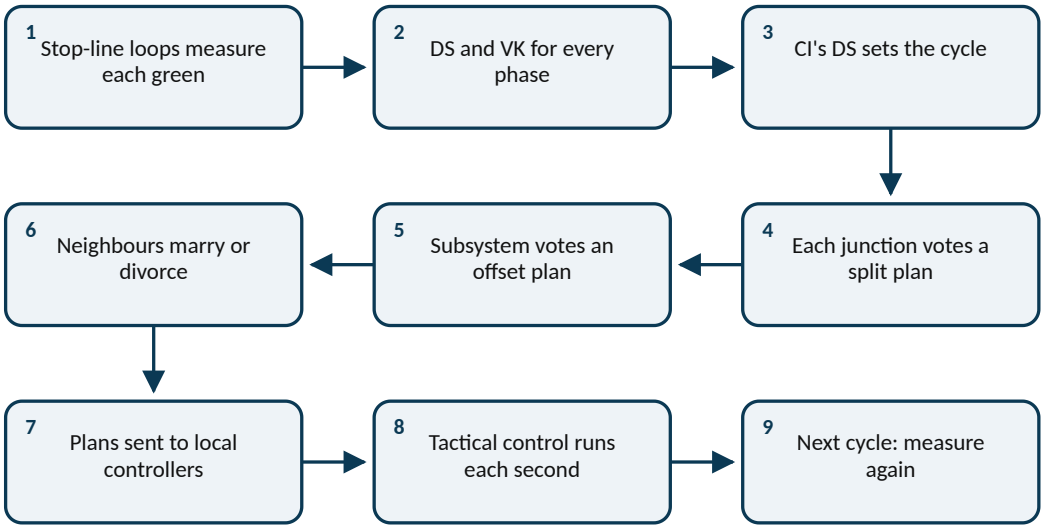


Figure 1. The strategic loop, once per cycle. What to notice: everything starts from DS and VK measured at stop-line loops; the cycle is decided first, because splits and offsets are fractions of it. Read left to right, then down.

TRAP

The CI is not 'the junction in the middle' or 'the first one'. It is the one whose demand limits the cycle, usually the busiest. Picking a quiet CI makes the whole subsystem's cycle too short.

PICTURE THIS · STOP 1 OF THE ROUTE

At the wharf, a team of up to ten little ferries is tied together with one rope. The biggest, puffing ferry at the front wears a captain's hat and bangs a drum; all the others row to its beat.

What it encodes: The roped ferries are a subsystem (1 to 10 signals); the captain is the critical intersection; the drumbeat is the shared cycle.

TRY IT NOW (about 90 seconds)

A junction had 40 s of green, 18 s of unoccupied loop time and 20 vehicles (h = 1.6 s). What is its DS?

Answer in the key at the back.

RECAP SO FAR

1. **Subsystems and CI: 1-10 signals share a cycle; the busiest (critical) intersection's DS sets it; others keep their own splits and offsets.**

CONCEPT 2 OF 8 · SCATS ADVANCED

Cycle length: bounded steps towards a target DS

HOOK

If the CI is overloaded, why not double the cycle straight away? SCATS deliberately refuses. It walks towards the right cycle a few seconds at a time.



IN PLAIN WORDS

Steering a big ship: you see the harbour entrance, but you turn the wheel a little each minute rather than spinning it. A small, steady correction is safer than one big turn based on one reading. *Where the picture breaks:* a ship's target stays put; the 'right' cycle keeps moving as traffic changes, which is another reason to move in small steps.

If the CI ran at DS with cycle C, the cycle that would have given the **target DS** (0.9 here) is roughly C × DS ÷ 0.9. That is the **desired cycle**.

SCATS never jumps there. It moves at most **6 s up** per cycle (**9 s** when DS is 1 or more, because the junction is overloaded) and at most **6 s down**. The result is kept between **LMIN** (40 s) and **LMAX** (140 s).

Above the **stretch cycle** (110 s here) extra time is not shared by the split plan; it all goes to one nominated phase, usually the main road, because long cycles are mainly for the heaviest movement.

$$\text{desired} = C \times DS \div DS_{target}$$

DS_{target} = 0.9 in this lab

$$C_{next} = C + \text{clamp}(\text{desired} - C, -6, +6 \text{ or } +9)$$

+9 when DS ≥ 1; then kept within 40 to 140 s

SIMPLE CALCULATION · Two cycles of a rising peak

The CI runs an 80 s cycle and measures DS = 1.05. Target 0.9.

1. Desired = 80 × 1.05 ÷ 0.9 = **93.3 s**.
2. Gap = 93.3 − 80 = 13.3 s. DS ≥ 1, so the step limit is 9 s.
3. Next cycle = 80 + 9 = **89 s**.
4. Next DS = 0.97: desired = 89 × 0.97 ÷ 0.9 = **95.9 s**; gap 6.9 s; limit 6 s (DS < 1).
5. Next cycle = 89 + 6 = **95 s**.

In two cycles SCATS moved from 80 s to 95 s, never by more than its step limit.

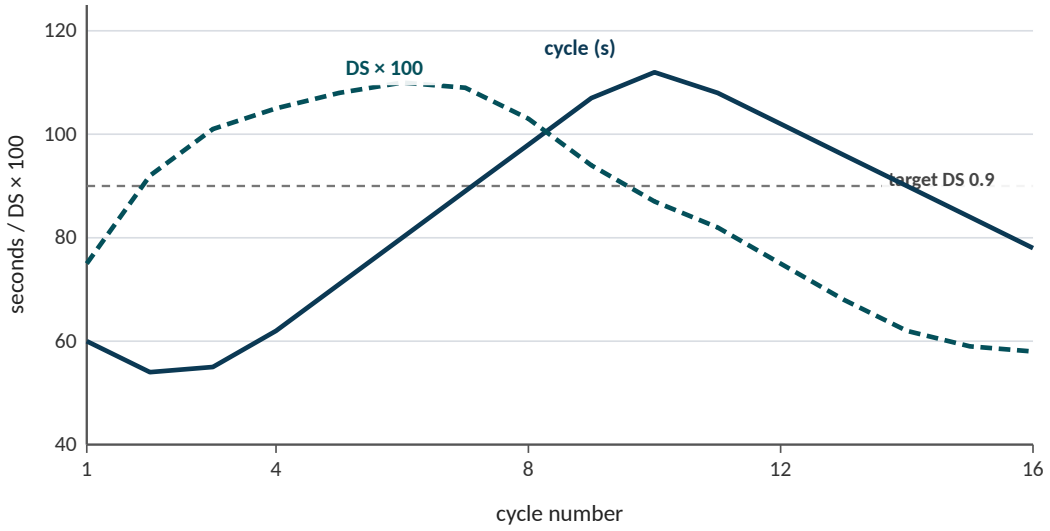


Figure 2. A simulated rush hour at the CI. What to notice: the cycle (top) climbs in steps of 6 or 9 s while DS is high, peaks at 112 s, then steps down by 6 s per cycle as DS falls. DS rises above 1 only briefly. Computed with this lab's next_cycle_length() on an invented demand pattern.

TRAP

The step limit is applied to the **change**, not to the cycle. 'Up 6 s' means $C_{next} = C + 6$ at most, not $C_{next} = 6$. And a low DS does not crash the cycle to LMIN; it walks down 6 s at a time.

PICTURE THIS · STOP 2 OF THE ROUTE

Inside the opera house a conductor speeds up the orchestra, but a rope on his baton lets him move it only six beats faster at a time, nine if the audience is stampeding. The tempo dial runs from 40 to 140, and a big '0.9' glows above the stage.

What it encodes: Tempo = cycle length; the rope = step limits 6 s (9 s if DS ≥ 1); the dial = LMIN 40 s and LMAX 140 s; 0.9 = target DS.

TRY IT NOW (about 90 seconds)

The cycle is 120 s and the CI's DS is 0.80. What is the next cycle? How much of it is 'stretch' time? *Answer in the key at the back.*

RECAP SO FAR

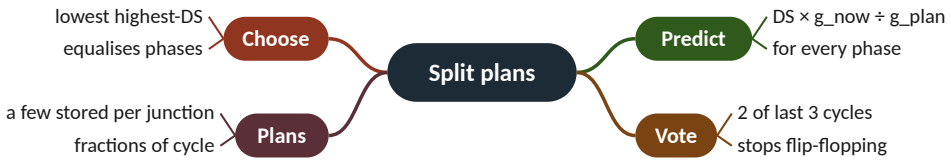
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2. **Cycle length: desired = C × DS ÷ 0.9; move at most +6 (+9 if DS ≥ 1) or −6 s per cycle, within 40-140 s; above 110 s extra goes to the main road.**

CONCEPT 3 OF 8 · SCATS ADVANCED

Split plans: equalise DS, then vote

HOOK

A junction has three phases. One is overloaded, one is half-empty. SCATS fixes this not by inventing new timings but by choosing among a few stored plans, and only after a vote.



IN PLAIN WORDS

Sharing pizzas at a party: you look at which table emptied its plate fastest and which left slices, then pick the seating plan that leaves every table equally full. You only move people if the same plan looks best twice out of three times. *Where the picture breaks:* SCATS can only pick from the few plans stored for that junction; it cannot invent a brand-new split.

Each junction stores a few **split plans**, each a share of the cycle for every phase. To compare them, SCATS asks: if this phase had had green g_{plan} instead of g_{now}, what would its DS have been? The traffic used DS × g_{now} seconds of green, so the predicted DS is that divided by g_{plan}.

The best plan is the one whose **highest** predicted DS is lowest. That choice equalises the phases: no phase is left badly overloaded while another has spare green.

A plan is only put into service after it wins consistently, here in **2 of the last 3** cycles. This voting smooths out one-off surges.

$$DS_{plan} = DS \times g_{now} \div g_{plan}$$

predicted DS of one phase under another plan

$$\text{best plan} = \text{the one with the smallest } \max(DS_{plan})$$

then it must win 2 of the last 3 votes

SIMPLE CALCULATION · Choosing among three plans

Greens now (plan SP1): A 40 s, B 15 s, C 20 s. Measured DS: A 0.95, B 0.50, C 0.70. SP2 gives 45/12/18 s, SP3 gives 35/15/25 s.

1. SP1 (no change): 0.95, 0.50, 0.70. Highest **0.95**.
2. SP2: A 0.95 × 40 ÷ 45 = 0.84; B 0.50 × 15 ÷ 12 = 0.63; C 0.70 × 20 ÷ 18 = 0.78. Highest **0.84**.
3. SP3: A 0.95 × 40 ÷ 35 = 1.09; B 0.50; C 0.70 × 20 ÷ 25 = 0.56. Highest **1.09**.
4. Smallest 'highest' is SP2 (0.84). SP2 gets this cycle's vote.
5. If SP2 also wins one of the next two cycles, it goes into service.

SP2 takes 5 s from the quiet side road and turn and gives them to the overloaded main road, bringing the worst DS down from 0.95 to 0.84.

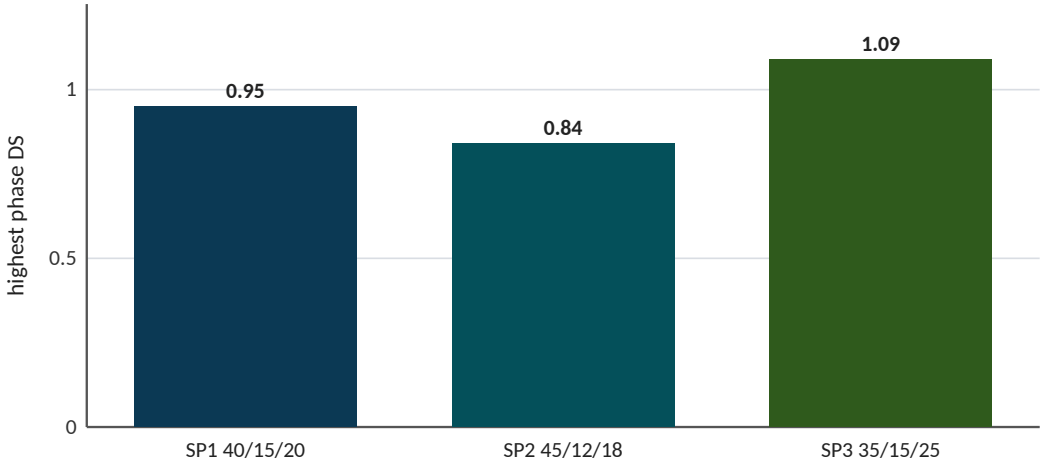


Figure 3. Highest predicted DS under each plan. What to notice: the winning plan is the shortest bar, because it leaves the worst phase least overloaded. Values from the worked example.

TRAP

Do not pick the plan with the lowest **average** DS. SP3 has a fine average but pushes phase A above 1. SCATS minimises the **worst** phase.

PICTURE THIS · STOP 3 OF THE ROUTE

At the fish market a stallholder balances three trays of fish on a giant see-saw scale, swapping fish until no tray sags. Before she nails the trays down, three seagulls vote; she needs two squawks for the same tray layout.

What it encodes: The trays are phases; balancing = minimising the highest DS; the three seagulls and two squawks = 2-of-3 voting.

TRY IT NOW (about 90 seconds)

Two-phase junction, greens now A 30 s and B 30 s, DS A 0.98 and B 0.60. Plan P gives A 36 s, B 24 s. What are the predicted DS values? Is P better? *Answer in the key at the back.*

RECAP SO FAR

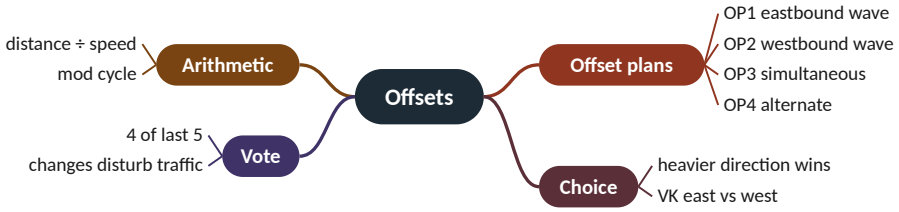
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3. **Split plans: predict DS × g_{now}/g_{plan} per phase; choose the plan with the lowest worst DS; adopt it after 2 of 3 wins.**

CONCEPT 4 OF 8 · SCATS ADVANCED

Offsets and green waves

HOOK

Drive along a well-timed road and every light turns green just as you arrive. That 'green wave' is just arithmetic: distance, speed and one subtraction.



IN PLAIN WORDS

A Mexican wave in a stadium: each section stands up a moment after its neighbour, timed so the wave travels at a steady speed. Time the lights the same way, and a platoon of cars rides the wave of greens. *Where the picture breaks:* a stadium wave goes one way; on a two-way road a perfect wave in one direction usually spoils the other, so SCATS must choose.

An **offset** is when a junction's green starts compared with the reference junction (the CI). For an eastbound green wave, each junction starts its green later by the travel time from the reference: distance ÷ progression speed. The result is taken **mod the cycle** (wrapped round, like a clock).

SCATS keeps a few **offset plans**: OP1 eastbound wave, OP2 westbound wave, OP3 all start together, OP4 alternate (every second junction shifted by half a cycle). It picks one from the directional flows: the clearly heavier direction (here, more than 1.3 times the other) gets the wave; balanced traffic gets OP3 or OP4.

Changing offsets disturbs traffic while the junctions shift, so offset plans need stronger agreement than splits: **4 of the last 5** cycles.

$$\text{OP1: } \text{offset}_i = (x_i - x_{ref}) \div v \pmod{C}$$

eastbound wave; OP2 uses the minus sign

$$\text{OP4: offset} = C \div 2 \text{ at every second junction}$$

two-way progression when travel time ≈ C/2

SIMPLE CALCULATION · An eastbound wave on four junctions

Junctions every 300 m, progression speed 13.9 m/s (50 km/h), cycle 80 s. VK eastbound 20, westbound 12.

1. Travel time per link: 300 ÷ 13.9 = **21.6 s**.
2. OP1 offsets: 0, 21.6, 43.2, 64.7 s (all under 80, no wrapping).
3. OP2 offsets: 0, −21.6, −43.2, −64.7 → mod 80 = 0, **58.4, 36.8, 15.3 s**.
4. Is east clearly heavier? 1.3 × 12 = 15.6, and 20 > 15.6, so **yes**.
5. This cycle votes OP1; it is adopted once it wins 4 of the last 5 votes.

Eastbound drivers at 50 km/h now meet green at every junction; westbound drivers get whatever is left.

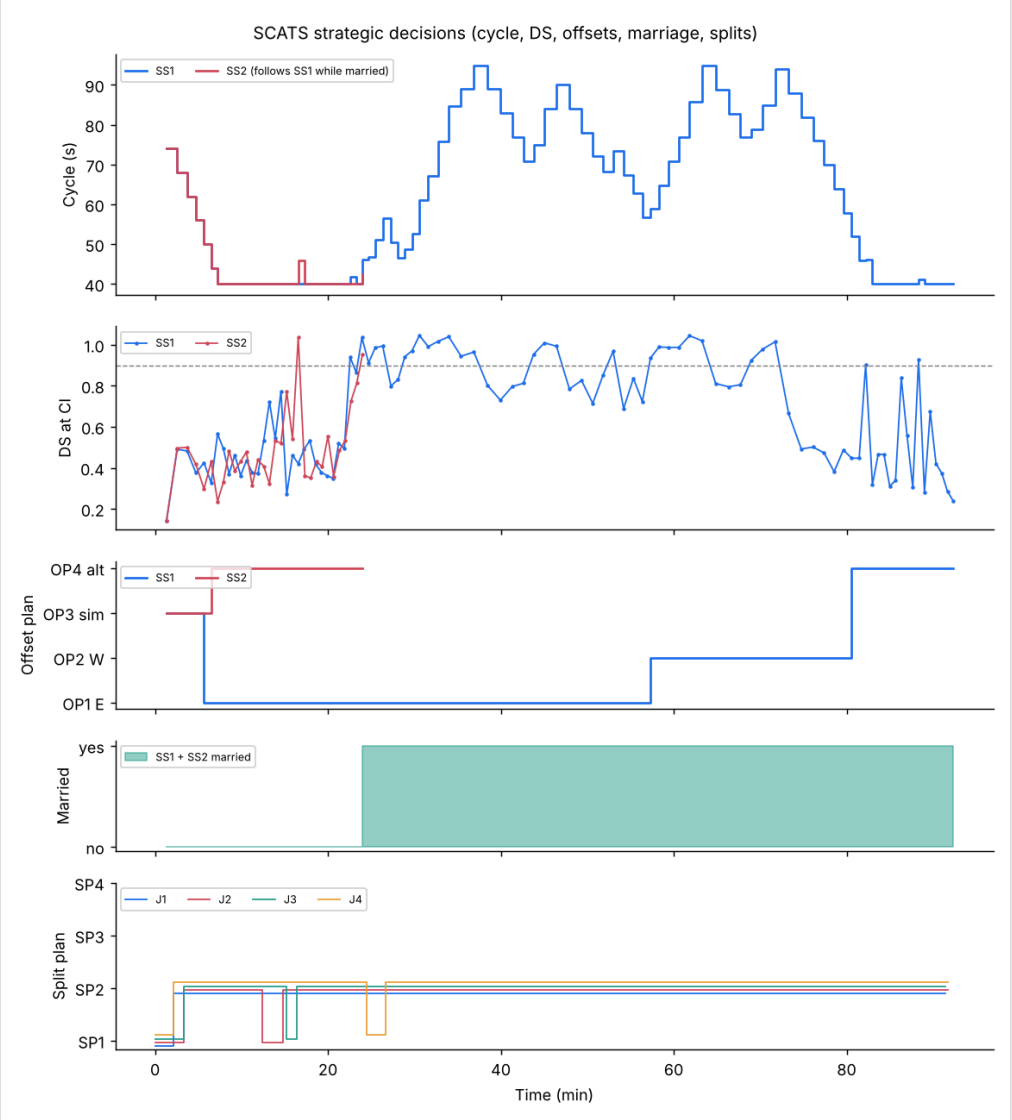


Figure 4. Strategic decisions on the SUMO corridor (Lab A1). What to notice: the cycle rises with each peak, the offset plan switches from eastbound (OP1) to westbound (OP2) progression and then to alternate (OP4) as traffic balances, and the two subsystems marry as the first peak builds.

TRAP

Offsets wrap round the cycle. A westbound offset of −21.6 s on an 80 s cycle is the same as +58.4 s. Forgetting 'mod C' gives negative start times that a controller cannot use.

PICTURE THIS · STOP 4 OF THE ROUTE

Along the harbour bridge, giant flags pop up one after another, left to right, exactly as a speeding red car passes each one. A stern referee holds up five cards and needs four to agree before the flags may change direction.

What it encodes: Flags popping in sequence = offsets making a green wave; the car's speed = progression speed; four of five cards = 4-of-5 offset voting.

TRY IT NOW (about 90 seconds)

Same corridor, but the cycle is 60 s. What are the OP1 offsets for junctions 1 to 4? *Answer in the key at the back.*

RECAP SO FAR

1. **Subsystems and CI:** 1-10 signals share a cycle; the busiest (critical) intersection's DS sets it; others keep their own splits and offsets.
2. **Cycle length:** desired = C × DS ÷ 0.9; move at most +6 (+9 if DS ≥ 1) or −6 s per cycle, within 40-140 s; above 110 s extra goes to the main road.
3. **Split plans:** predict DS × g_{now}/g_{plan} per phase; choose the plan with the lowest worst DS; adopt it after 2 of 3 wins.
4. **Offsets: offset = distance ÷ speed (mod C); OP1/OP2 waves, OP3 together, OP4 alternate; heavier direction (>1.3×) wins; adopt after 4 of 5 votes.**

CONCEPT 5 OF 8 · SCATS ADVANCED

Marriage and divorce

HOOK

Two neighbouring subsystems each run their own cycle. When lots of traffic flows between them, they would do better as one. When it fades, they would do better apart. How do you stop them changing their minds every minute?



IN PLAIN WORDS

A thermostat that turns heating on at 18 °C and off at 21 °C, not at the same temperature. The gap stops the boiler clicking on and off. SCATS uses a similar gap, plus three agreeing readings in a row. *Where the picture breaks:* a thermostat acts on one reading; SCATS waits for three consecutive votes, which a thermostat does not.

Two adjacent subsystems can be **married**: they run on a common cycle (the **larger** of the two they each need) and share a master clock, so the offset between them means something. When the traffic between them drops, coordination no longer pays for the extra cycle time, and they **divorce**.

The decision uses the link flow (VK) between them. In this lab they marry when link VK is at least **18** and the two cycles are within 25 s of each other; once married they stay married until VK falls below **10**. Different thresholds for on and off are called **hysteresis**.

A change also needs **3 consecutive votes**. One busy cycle cannot marry them, and one quiet cycle cannot divorce them.

$$\text{marry if VK} \geq 18 \ (\text{and } |C_A - C_B| \leq 25 \text{ s}) \text{ for 3 cycles in a row}$$

then both run $\max(C_A, C_B)$

$$\text{divorce if VK} < 10 \text{ for 3 cycles in a row}$$

between 10 and 18 nothing changes

SIMPLE CALCULATION · Tracing link VK cycle by cycle

Divorced subsystems with cycles 80 s and 90 s. Link VK per cycle: 12, 15, 19, 20, 17, 21, 22, 23, 16, 12, 9, 11, 8, 7, 6.

1. Cycles 3-4 (19, 20) vote 'marry'; cycle 5 (17) breaks the run, count back to 0.
2. Cycles 6, 7, 8 (21, 22, 23): three in a row. **Married at cycle 8**, common cycle 90 s.
3. Cycles 9-10 (16, 12) are above 10: stay married.
4. Cycle 11 (9) votes 'divorce'; cycle 12 (11) breaks the run.
5. Cycles 13, 14, 15 (8, 7, 6): three in a row. **Divorced at cycle 15**.

Two state changes in fifteen noisy cycles, instead of the many flips a single threshold would give.

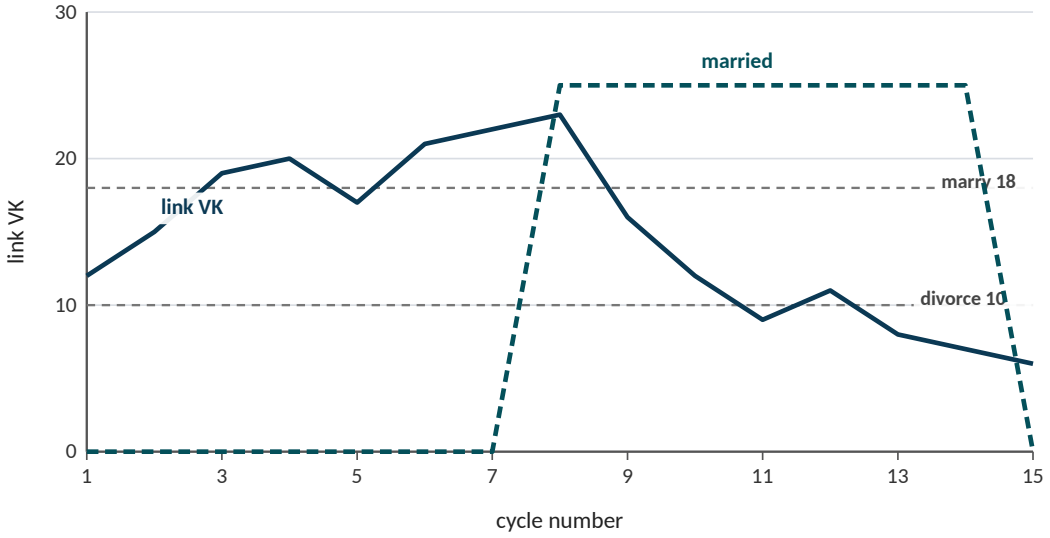


Figure 5. Link VK and the marriage state. What to notice: the state (married = 25 on this scale) switches on only after three cycles at or above 18, and off only after three below 10. Readings between the dashed lines never change anything. Computed with this lab's MarriageLogic.

TRAP

Married subsystems do not average their cycles; they both take the **larger** one, so the busier subsystem is not starved. That is also the cost of marriage: the quieter one runs a longer cycle than it needs.

PICTURE THIS · STOP 5 OF THE ROUTE

On the rocks above the harbour, two ferries get married in a tiny chapel. The priest needs 18 roses to start the wedding but only takes the rings back if fewer than 10 remain, and three bells must ring in a row each time. The couple share the bigger of their two clocks.

What it encodes: Marry at VK 18, divorce below 10 (hysteresis); three bells = 3 consecutive votes; the bigger clock = the larger cycle.

TRY IT NOW (about 90 seconds)

Divorced, cycles 70 s and 100 s, link VK 25, 26, 27. Do they marry? *Answer in the key at the back.*

RECAP SO FAR

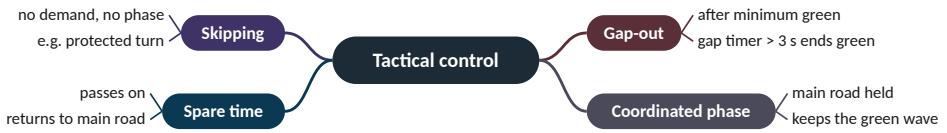
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3. **Split plans:** predict DS × g_{now}/g_{plan} per phase; choose the plan with the lowest worst DS; adopt it after 2 of 3 wins.
4. **Offsets:** offset = distance ÷ speed (mod C); OP1/OP2 waves, OP3 together, OP4 alternate; heavier direction (>1.3×) wins; adopt after 4 of 5 votes.
5. **Marriage: neighbours share the larger cycle when link VK ≥ 18 for 3 cycles (cycles within 25 s); divorce when VK < 10 for 3 cycles.**

CONCEPT 6 OF 8 · SCATS ADVANCED

Tactical control: gap-out, skipping, holding

HOOK

The regional computer plans greens once per cycle. But what if the side road empties after 8 seconds of a planned 25? Something has to notice, every second.



IN PLAIN WORDS

A lift door: it stays open while people keep walking in, and closes once nobody has come through for a few seconds. If nobody pressed the button on a floor, the lift does not stop there at all. *Where the picture breaks:* a lift has no neighbours to coordinate with; SCATS holds the main-road green on purpose so it does not break the green wave.

Strategic control gives each phase a **planned green**. The local controller then checks, every second, whether that green is still useful. After the **minimum green**, a **gap timer** restarts each time a vehicle crosses the loop. If it reaches the gap setting (3 s here), the phase **gaps out** and its unused time passes to the next phase.

Demand-dependent phases, such as a protected right-turn arrow, are **skipped** when nobody is waiting.

The **coordinated phase** (the main road) is normally **held** rather than gapped out, because ending it early would break the green wave. Spare time from the other phases returns to it.

$$\text{end green when } t \geq g_{\min} \text{ and } (t - \text{last vehicle}) \geq \text{gap setting}$$

gap setting 3 s here; never beyond the planned or maximum green

SIMPLE CALCULATION · Does this side-road green gap out?

Planned green 25 s, minimum green 8 s, gap setting 3 s. Vehicles cross the loop at 1.0, 2.5, 4.0 and 5.5 s, then the next one would come at 9.5 s.

1. Before 8 s the green cannot end (minimum green).
2. At 8.0 s the gap since the last car (5.5 s) is 2.5 s, below 3 s: keep going.
3. At 8.5 s the gap is 3.0 s: **gap-out**. The green ends at 8.5 s.
4. Unused time: 25 − 8.5 = **16.5 s**, passed on to the next phase.
5. The car arriving at 9.5 s waits for the next cycle.

Instead of 16.5 s of empty green, the main road gets that time. This is why switching off tactical control hurt most in the SUMO ablation.

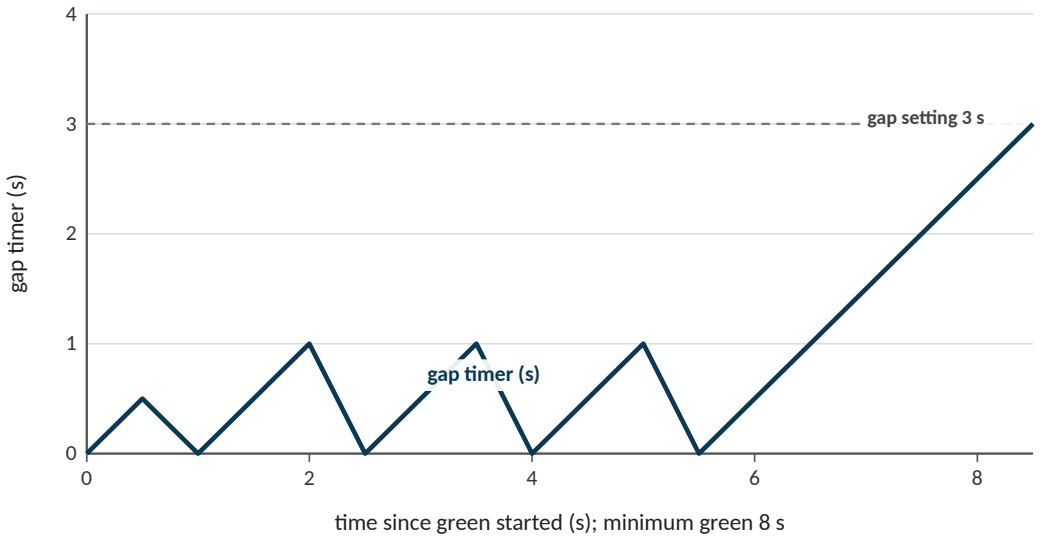


Figure 6. The gap timer during one green. What to notice: the timer resets to 0 at every vehicle; the green may end only after the minimum green, the first time the timer reaches the 3 s line (8.5 s). Values from the worked example.

TRAP

Gap-out does not apply to every phase. The coordinated main-road phase is held so the green wave survives, and no phase can gap out before its minimum green.

PICTURE THIS · STOP 6 OF THE ROUTE

In the lighthouse, a lift door stays open while tourists stream in, then slams after three silent seconds, but never before an eight-second bell. The lift skips floors where nobody pressed a button. On the main-road floor a guard props the door open with a giant anchor.

What it encodes: Three silent seconds = gap setting 3 s; eight-second bell = minimum green; skipped floors = phase skipping; anchor = the held coordinated phase.

TRY IT NOW (about 90 seconds)

Minimum green 6 s, gap 3 s, vehicles at 0.5, 2.0 and 6.5 s, then none. When does the green end?

Answer in the key at the back.

RECAP SO FAR

1. **Subsystems and CI:** 1-10 signals share a cycle; the busiest (critical) intersection's DS sets it; others keep their own splits and offsets.
2. **Cycle length:** desired = C × DS ÷ 0.9; move at most +6 (+9 if DS ≥ 1) or −6 s per cycle, within 40-140 s; above 110 s extra goes to the main road.
3. **Split plans:** predict DS × g_{now}/g_{plan} per phase; choose the plan with the lowest worst DS; adopt it after 2 of 3 wins.
4. **Offsets:** offset = distance ÷ speed (mod C); OP1/OP2 waves, OP3 together, OP4 alternate; heavier direction (>1.3×) wins; adopt after 4 of 5 votes.
5. **Marriage:** neighbours share the larger cycle when link VK ≥ 18 for 3 cycles (cycles within 25 s); divorce when VK < 10 for 3 cycles.
6. **Tactical: after minimum green a 3 s gap ends a green early; empty demand phases are skipped; the main road is held; spare time passes on.**

CONCEPT 7 OF 8 · SCATS ADVANCED

Public transport priority

HOOK

A bus with 60 people arrives a second after its green ends. Should 60 people wait for the next cycle so that 5 cars on the side road get their turn?



IN PLAIN WORDS

At a busy shop counter, a customer in a wheelchair (high priority) is served next, someone with one item may be squeezed in early (medium), and everyone else waits their turn (low). *Where the picture breaks:* a shop has one queue; at a junction every second given to the bus is taken from someone else's green, so priority has a cost.

SCATS can receive **priority requests** from buses and trams, for example from a transponder or a detector in a bus lane. Three levels are commonly described.

High priority is a 'hurry call': the transit phase is served as soon as it is safe, skipping other phases if needed. It is typical for trams and emergency vehicles. **Medium** priority is a flexible window: the current green is extended, or other phases are shortened, so the transit phase comes sooner. It is typical for buses. **Low** priority means the vehicle waits in normal sequence.

In Lab A1 buses ran every 4 minutes each way. Mean bus delay was **110.8 s** with fixed time, **51.8 s** actuated and **29.7 s** with SCATS and bus priority.

SIMPLE CALCULATION · A bus that just missed its green

Cycle 80 s: phase A (bus) green 0-40 s, intergreen 5 s, phase B green 45-75 s, intergreen 5 s. The bus arrives at 42 s. Phase B has a 10 s minimum green.

1. **Low:** wait for the next A green at 80 s. Delay = 80 − 42 = **38 s**.
2. **Medium:** cut B to its 10 s minimum: B ends at 55 s, intergreen to 60 s. Delay = 60 − 42 = **18 s**.
3. **High:** skip B entirely; A returns after the intergreen at 45 s. Delay = 45 − 42 = **3 s**.
4. Cost: medium takes 20 s from B; high takes all 30 s.

Higher priority saves the bus more time but costs the side road more. That is why buses usually get medium, not high.

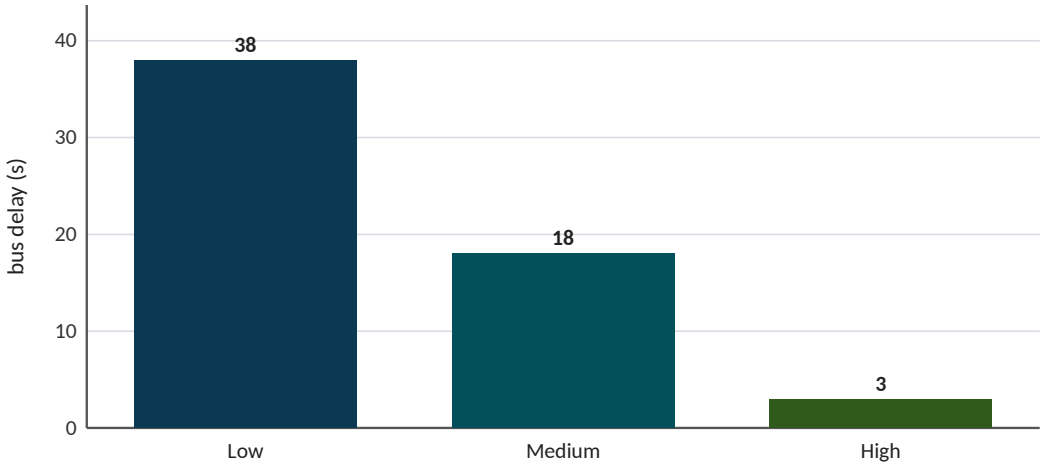


Figure 7. Bus delay at each priority level (worked example). What to notice: each step up in priority cuts the bus delay sharply, but the time comes from phase B. Illustrative timings from the calculation, not SUMO.

TRAP

Priority is not free. In the ablation, switching bus priority off raised **bus** delay from 29.7 to 46.4 s but also raised mean delay for everyone from 36.9 to 42.5 s here, because buses ran on the main road with the green wave. On another corridor, aggressive priority can raise general delay instead.

PICTURE THIS · STOP 7 OF THE ROUTE

At the tram stop three vehicles queue for a magic gate. A fire-red tram blasts a hurry horn and the gate flies open at once. A yellow bus waves a ticket and the gate swings early. A grey van just waits its turn, tapping the wheel.

What it encodes: Hurry horn = high priority (serve at once); waved ticket = medium (extend or start early); waiting van = low (normal sequence).

TRY IT NOW (about 90 seconds)

Same junction, but the bus arrives at 38 s, before A's green ends. Under medium priority, what happens and what is the bus delay? *Answer in the key at the back.*

RECAP SO FAR

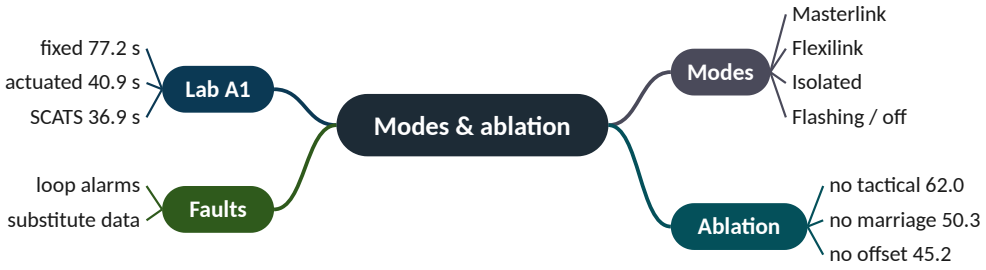
1. **Subsystems and CI:** 1-10 signals share a cycle; the busiest (critical) intersection's DS sets it; others keep their own splits and offsets.
2. **Cycle length:** desired = C × DS ÷ 0.9; move at most +6 (+9 if DS ≥ 1) or −6 s per cycle, within 40-140 s; above 110 s extra goes to the main road.
3. **Split plans:** predict DS × g_{now}/g_{plan} per phase; choose the plan with the lowest worst DS; adopt it after 2 of 3 wins.
4. **Offsets:** offset = distance ÷ speed (mod C); OP1/OP2 waves, OP3 together, OP4 alternate; heavier direction (>1.3×) wins; adopt after 4 of 5 votes.
5. **Marriage:** neighbours share the larger cycle when link VK ≥ 18 for 3 cycles (cycles within 25 s); divorce when VK < 10 for 3 cycles.
6. **Tactical:** after minimum green a 3 s gap ends a green early; empty demand phases are skipped; the main road is held; spare time passes on.
7. **Bus priority: high = hurry call, medium = extend or start early, low = wait; Lab A1 bus delay 110.8 / 51.8 / 29.7 s.**

CONCEPT 8 OF 8 · SCATS ADVANCED

Operating modes and the full system on SUMO

HOOK

What happens to a city's signals when the cable to the control centre is cut? And if you remove one SCATS mechanism at a time, which one does the corridor miss most?



IN PLAIN WORDS

An orchestra with a conductor (Masterlink); the same orchestra following printed sheet music and a shared metronome when the conductor leaves (Flexilink); musicians each playing by ear alone (Isolated). *Where the picture breaks:* musicians notice a missing conductor at once; SCATS needs alarms and configuration to detect faulty loops and drop down a mode.

Masterlink is full adaptive operation: strategic plans from the regional computer, tactical control locally. **Flexilink** is the fallback: time-of-day plans with synchronised clocks, so offsets still hold, with local actuation inside them. **Isolated** means local vehicle actuation only, with no coordination. Last comes **flashing or off**, for faults and power loss.

SCATS also raises alarms for implausible loops (no counts for many cycles, or the loop stuck 'on'), and engineers configure substitutes. The Mastery level tests this.

Lab A1 ran everything on four 3-phase junctions 300 m apart, in two subsystems that may marry, for 90 minutes with tidal peaks and buses every 4 minutes. Lab A2, the **ablation**, switched one mechanism off at a time.

$$\text{change} = (\text{delay without mechanism} - \text{delay with all}) \div \text{delay with all}$$

how much the corridor misses each mechanism

SIMPLE CALCULATION · Reading the ablation

Full SCATS: 36.94 s mean delay. With one mechanism off: no tactical 62.04, no marriage 50.33, no offset 45.22, no bus priority 42.49, no split 37.50, no cycle 34.42 s.

1. No tactical: (62.04 − 36.94) ÷ 36.94 = **+68%**.
2. No marriage: 13.39 ÷ 36.94 = **+36%**. No offset: 8.28 ÷ 36.94 = **+22%**.
3. No bus priority: 5.55 ÷ 36.94 = **+15%**. No split: 0.56 ÷ 36.94 = **+1.5%**.
4. No cycle adaptation: −2.52 ÷ 36.94 = **−6.8%** (slightly better).
5. Versus fixed time (77.2 s): SCATS cuts delay by **52%**; versus actuated (40.9 s) by **10%**.

On this corridor, second-by-second tactical control and coordination (marriage, offsets) carry most of the benefit. An 80 s fixed cycle happened to suit this demand, so cycle adaptation did not pay here.

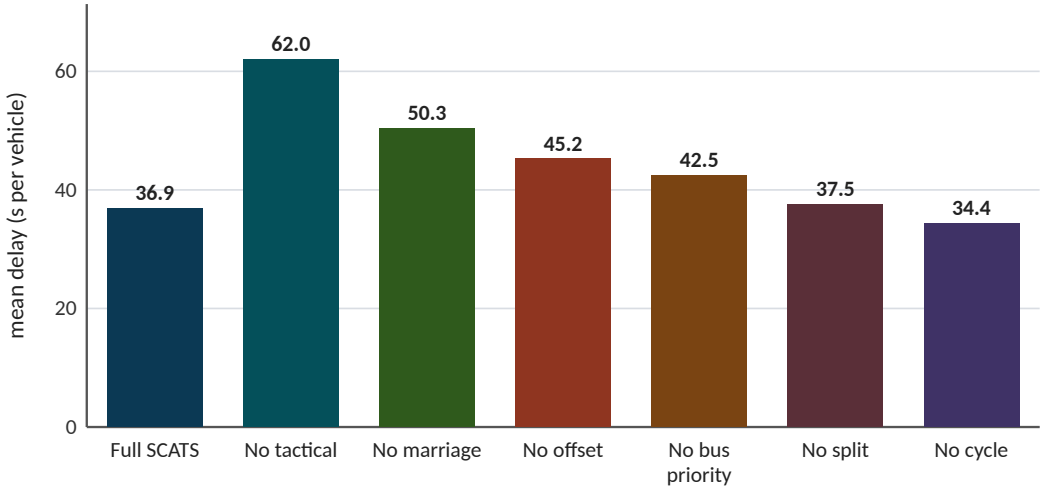


Figure 8. Lab A2 ablation: mean delay with one mechanism switched off (SUMO, seed 1). What to notice: the tallest bar shows the mechanism the corridor misses most. 'No cycle' is below 'Full': a fixed 80 s cycle happened to suit this demand.

TRAP

An ablation result is about **this** corridor and **this** demand. 'No cycle adaptation was better' does not mean cycle adaptation is useless; it pays off when demand swings far from the fixed value.

PICTURE THIS · STOP 8 OF THE ROUTE

In the control tower a conductor leads an orchestra of traffic lights. When she trips over the cable, the lights pick up sheet music and a ticking metronome. Behind her a scoreboard shows seven pillars; the tallest, 62, is labelled 'tactical'.

What it encodes: Conductor = Masterlink; sheet music + metronome = Flexilink; the scoreboard = ablation, no tactical 62.0 s versus 36.9 s full.

TRY IT NOW (about 90 seconds)

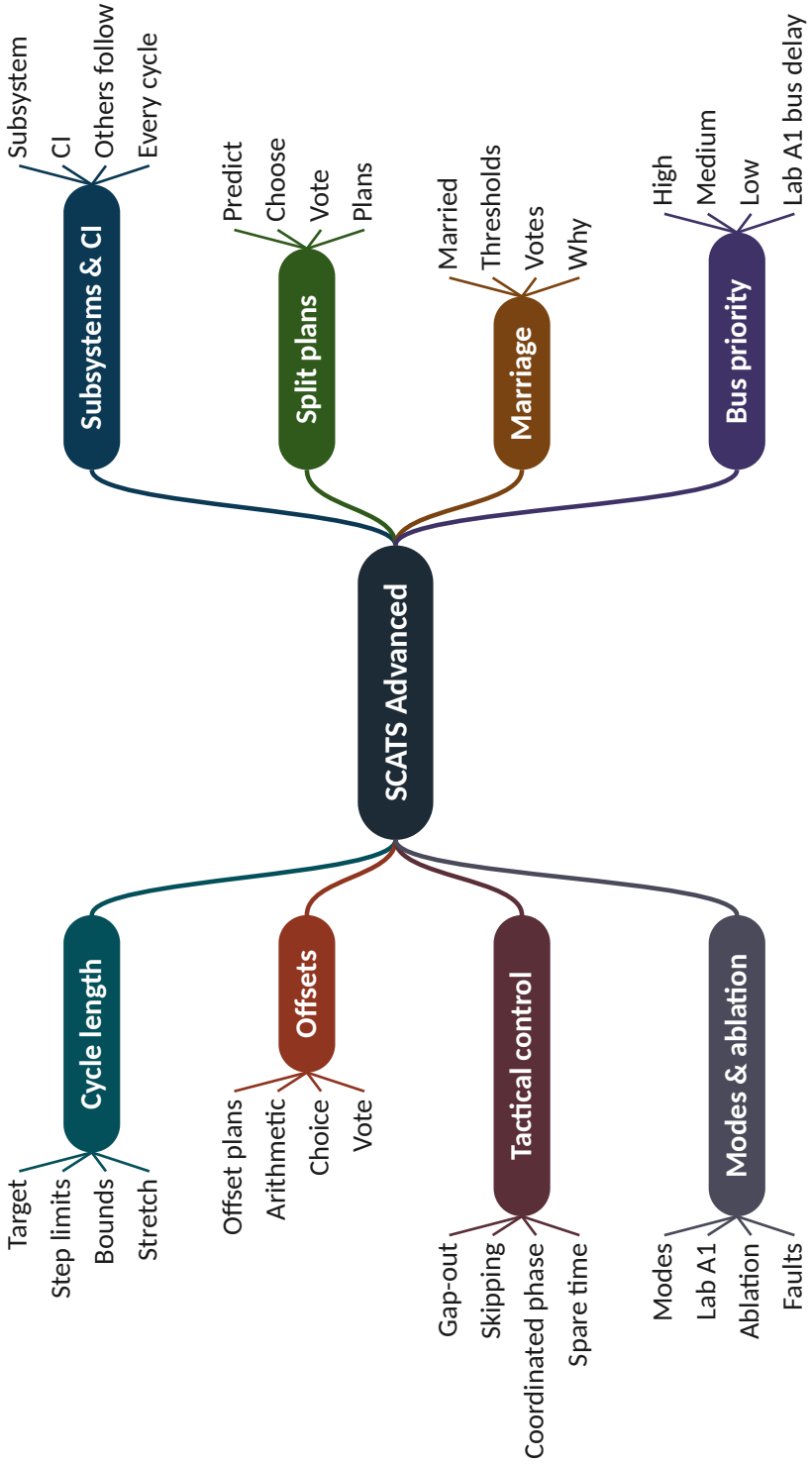
Lab A1 stops per vehicle were 1.82 (fixed), 1.57 (actuated) and 1.30 (SCATS). By what percentage did SCATS cut stops compared with fixed time? *Answer in the key at the back.*

RECAP SO FAR

1. **Subsystems and CI:** 1-10 signals share a cycle; the busiest (critical) intersection's DS sets it; others keep their own splits and offsets.
2. **Cycle length:** desired = C × DS ÷ 0.9; move at most +6 (+9 if DS ≥ 1) or −6 s per cycle, within 40-140 s; above 110 s extra goes to the main road.
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5. **Marriage:** neighbours share the larger cycle when link VK ≥ 18 for 3 cycles (cycles within 25 s); divorce when VK < 10 for 3 cycles.
6. **Tactical:** after minimum green a 3 s gap ends a green early; empty demand phases are skipped; the main road is held; spare time passes on.
7. **Bus priority:** high = hurry call, medium = extend or start early, low = wait; Lab A1 bus delay 110.8 / 51.8 / 29.7 s.
8. **Modes and ablation: Masterlink → Flexilink → Isolated → flashing; Lab A1 delay 77.2 / 40.9 / 36.9 s; ablation: tactical matters most (62.0 s off).**

The master mind map

Every concept and its main branches on one page. Turn the page sideways. Try redrawing it from memory on blank paper, then compare.



Connections and contrasts

Strategic versus tactical control

	Strategic	Tactical
Where it runs	regional computer	local controller at the junction
How often	once per cycle	every second
Decides	cycle, split plan, offset plan, marriage	gap-out, skipping, holding, priority
Damping	step limits, 2-of-3 and 4-of-5 votes, hysteresis	minimum green, gap setting
Ablation (delay when off)	marriage 50.3, offset 45.2, split 37.5, cycle 34.4 s	62.0 s

Things beginners mix up

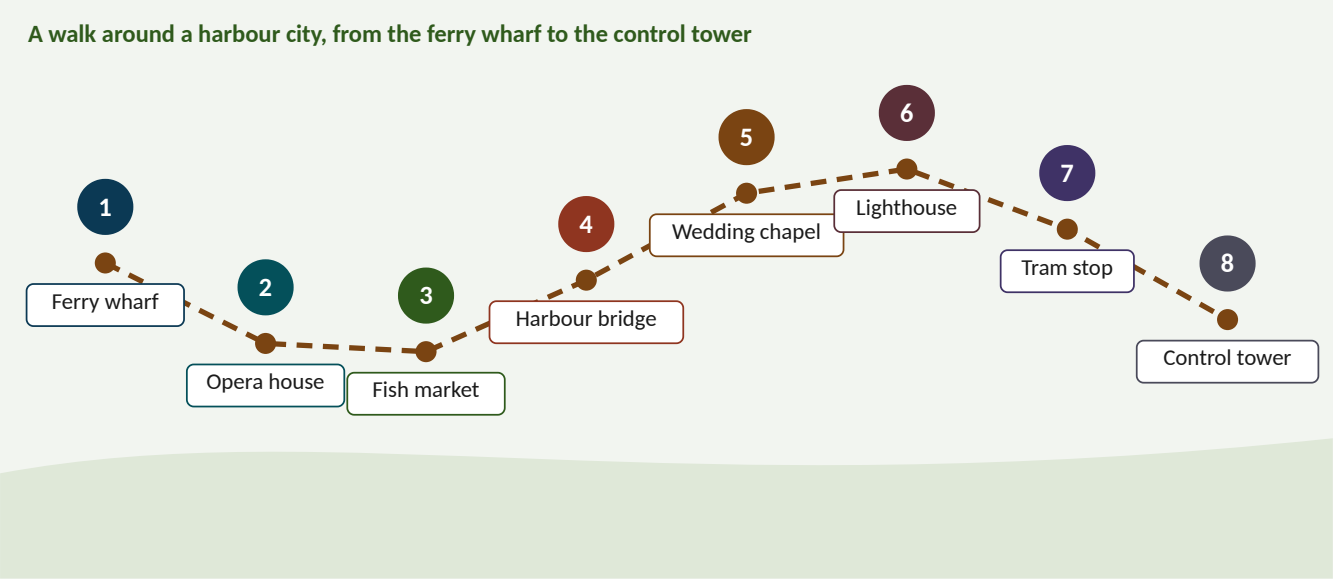
This…	…is not this	The one fact that tells them apart
Split voting (2 of 3)	**Offset voting (4 of 5)**	Offset changes disturb traffic more, so they need stronger agreement.
Married cycle	**Average of the two cycles**	Married subsystems both take the larger cycle.
Lowest worst-phase DS	**Lowest average DS**	SCATS picks the split plan that minimises the highest phase DS.
Gap-out	**Phase skipping**	Gap-out ends a green that started; skipping never starts a phase with no demand.
Flexilink	**Isolated**	Flexilink keeps synchronised clocks, so offsets still hold; Isolated has no coordination.

Symbols and terms

Symbol / term	Meaning
CI	critical intersection: its DS sets the subsystem cycle
DS	degree of saturation (g − (T − nh)) / g
VK	reconstituted flow, vehicles a green could carry
C, C_{next}	current and next cycle length (s)
LMIN, LMAX	cycle bounds, 40 and 140 s here
SP, OP	split plan, offset plan
mod C	wrap round the cycle, like a clock
g_{min}	minimum green before gap-out is allowed

The route: A walk around a harbour city, from the ferry wharf to the control tower

Picture a sunny harbour city. You start at the ferry wharf and walk around the water to the traffic control tower on the far headland. Eight places, one per concept. Make every scene loud, silly and physical.



The memory route. Walk it in order; each stop holds one concept.

Stop 1: Ferry wharf (Subsystems and the critical intersection). At the wharf, a team of up to ten little ferries is tied together with one rope. The biggest, puffing ferry at the front wears a captain's hat and bangs a drum; all the others row to its beat. *The roped ferries are a subsystem (1 to 10 signals); the captain is the critical intersection; the drumbeat is the shared cycle.*

Stop 2: Opera house (Cycle length: bounded steps towards a target DS). Inside the opera house a conductor speeds up the orchestra, but a rope on his baton lets him move it only six beats faster at a time, nine if the audience is stampeding. The tempo dial runs from 40 to 140, and a big '0.9' glows above the stage. *Tempo = cycle length; the rope = step limits 6 s (9 s if DS ≥ 1); the dial = LMIN 40 s and LMAX 140 s; 0.9 = target DS.*

Stop 3: Fish market (Split plans: equalise DS, then vote). At the fish market a stallholder balances three trays of fish on a giant see-saw scale, swapping fish until no tray sags. Before she nails the trays down, three seagulls vote; she needs two squawks for the same tray layout. *The trays are phases; balancing = minimising the highest DS; the three seagulls and two squawks = 2-of-3 voting.*

Stop 4: Harbour bridge (Offsets and green waves). Along the harbour bridge, giant flags pop up one after another, left to right, exactly as a speeding red car passes each one. A stern referee holds up five cards and needs four to agree before the flags may change direction. *Flags popping in sequence = offsets making a green wave; the car's speed = progression speed; four of five cards = 4-of-5 offset voting.*

Stop 5: Wedding chapel (Marriage and divorce). On the rocks above the harbour, two ferries get married in a tiny chapel. The priest needs 18 roses to start the wedding but only takes the rings back if fewer than 10 remain, and three bells must ring in a row each time. The couple share the bigger of their two clocks. *Marry at VK 18, divorce below 10 (hysteresis); three bells = 3 consecutive votes; the bigger clock = the larger cycle.*

Stop 6: Lighthouse (Tactical control: gap-out, skipping, holding). In the lighthouse, a lift door stays open while tourists stream in, then slams after three silent seconds, but never before an eight-second bell. The lift skips floors where nobody pressed a button. On the main-road floor a guard props the door open with a giant anchor. *Three silent seconds = gap setting 3 s; eight-second bell = minimum green; skipped floors = phase skipping; anchor = the held coordinated phase.*

Stop 7: Tram stop (Public transport priority). At the tram stop three vehicles queue for a magic gate. A fire-red tram blasts a hurry horn and the gate flies open at once. A yellow bus waves a ticket and the gate swings early. A grey van just waits its turn, tapping the wheel. *Hurry horn = high priority (serve at once); waved ticket = medium (extend or start early); waiting van = low (normal sequence).*

Stop 8: Control tower (Operating modes and the full system on SUMO). In the control tower a conductor leads an orchestra of traffic lights. When she trips over the cable, the lights pick up sheet music and a ticking metronome. Behind her a scoreboard shows seven pillars; the tallest, 62, is labelled 'tactical'. *Conductor = Masterlink; sheet music + metronome = Flexilink; the scoreboard = ablation, no tactical 62.0 s versus 36.9 s full.*

WALK-THROUGH DRILL

Close the guide. Walk the route in your head, saying each stop out loud and the one number or formula it holds. Then check against the cheat sheet. Repeat until you can do it in under two minutes.

Self-quiz

18 questions: warm-up (recall), core (use the ideas), challenge (combine them). Answer on paper first, then check the key.

Warm-up

Q1. Which intersection's DS sets a subsystem's cycle length?

A. The first one on the road
B. The critical intersection
C. The quietest one
D. The average of all

Q2. How many recent cycles must a split plan win before it goes into service in this lab?

A. 1 of 1
B. 2 of 3
C. 4 of 5
D. 3 in a row

Q3. When two subsystems marry, which cycle do they run?

A. The smaller
B. The average
C. The larger
D. A fixed 80 s

Q4. Which priority level is a 'hurry call'?

A. High
B. Medium
C. Low
D. None of them

Q5. Which operating mode keeps synchronised clocks without the regional computer?

A. Masterlink
B. Flexilink
C. Isolated
D. Flashing

Core

Q6. C = 100 s and the CI's DS = 1.20 (target 0.9). What is the next cycle?

A. 106 s
B. 109 s
C. 133 s
D. 140 s

Q7. C = 60 s and DS = 0.45. What is the next cycle?

A. 30 s
B. 40 s
C. 54 s
D. 60 s

Q8. A phase had DS 0.90 with 30 s of green. Under a plan giving it 36 s, its predicted DS is:

A. 0.75
B. 0.90
C. 1.08
D. 1.20

Q9. Plan X predicts DS 0.85/0.85/0.40; plan Y predicts 0.70/0.70/0.95. Which does SCATS prefer?

A. X, its worst phase is 0.85
B. Y, its average is lower
C. Y, two phases are lower
D. Neither, both above 0.9

Q10. Junctions 400 m apart, progression 16 m/s, cycle 90 s. OP1 offset of the third junction (800 m from the reference)?

A. 25 s
B. 40 s
C. 50 s
D. 65 s

Q11. VK east 14, west 13, balanced. Which offset plans can SCATS choose?

A. OP1 only
B. OP2 only
C. OP3 or OP4
D. Any; it picks at random

Q12. Married subsystems see link VK 15, 14, 16, 12. What happens?

A. They divorce after 3 cycles
B. They stay married
C. They divorce at once
D. They change cycle to the smaller

Q13. Minimum green 8 s, gap 3 s; last vehicle at 2 s, none after. When does the green end?

A. 5 s
B. 8 s
C. 11 s
D. At the planned green

Q14. Why is the main-road (coordinated) phase normally not allowed to gap out?

A. It has no detector
B. Ending it early would break the green wave
C. It always has demand
D. Rules forbid it

Challenge

Q15. A tram must be served within seconds, but the side road has a pedestrian phase running. What does high priority do?

A. Cuts the pedestrian phase instantly
B. Serves the tram as soon as safely possible, after minimums and intergreens
C. Waits for the next cycle
D. Turns all signals red

Q16. In the ablation, switching off cycle adaptation lowered delay. Best conclusion?

A. Cycle adaptation is useless
B. The fixed 80 s happened to suit this demand; test wider demand swings
C. SUMO is wrong
D. Remove it from SCATS

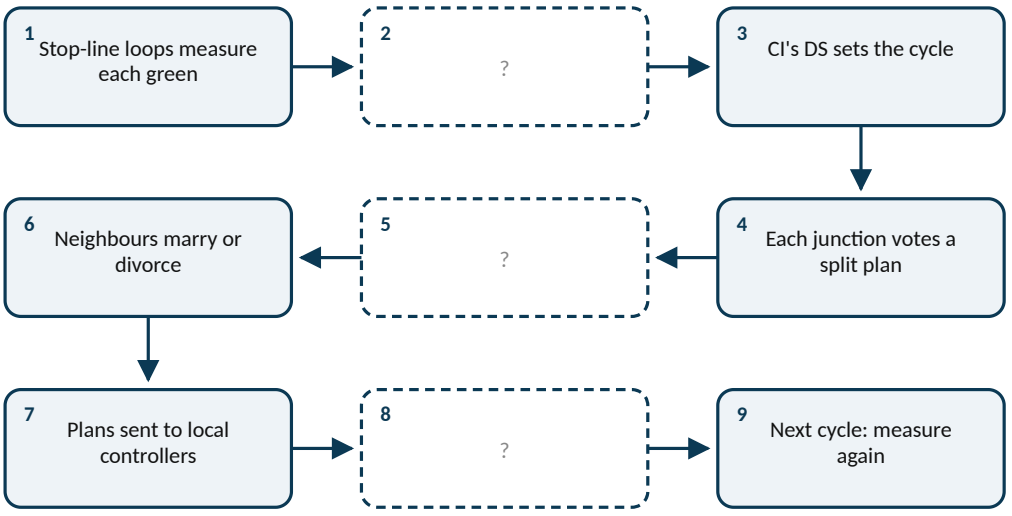
Q17. Two subsystems need 70 s and 110 s with link VK 30. Why might marriage still be refused?

A. VK is too low
B. Their cycles differ by more than 25 s
C. Marriage needs 4 of 5 votes
D. Only the CI can marry

Q18. Which pair of mechanisms carried most of SCATS's benefit on the Lab A2 corridor?

A. Split and cycle adaptation
B. Tactical control and marriage
C. Bus priority and cycle adaptation
D. Offset and split adaptation

Draw it. Fill the blank boxes of this flow from memory (it is Figure 1).



Where to go if you missed a question

Missed	Do this
Q1	Re-read concept 1: Subsystems and the critical intersection
Q6, Q7	Re-read concept 2: Cycle length: bounded steps towards a target DS
Q2, Q8, Q9	Re-read concept 3: Split plans: equalise DS, then vote
Q10, Q11	Re-read concept 4: Offsets and green waves
Q3, Q12, Q17	Re-read concept 5: Marriage and divorce
Q13, Q14	Re-read concept 6: Tactical control: gap-out, skipping, holding
Q4, Q15	Re-read concept 7: Public transport priority
Q5, Q16, Q18	Re-read concept 8: Operating modes and the full system on SUMO

Make it stick

BUILD THE SMALLEST WORKING EXAMPLE (15 min)

On paper, run a CI for six cycles. Start at C = 70 s. Use DS values 1.10, 1.05, 0.95, 0.90, 0.75, 0.70. Apply desired = C × DS ÷ 0.9 and the step limits (+9 if DS ≥ 1, +6, −6). Plot cycle against cycle number.

Self-check: You should get 70 → 79 → 88 → 92.9 → 92.9 → 86.9 → 80.9 s (the third step is only +4.9 s because desired = 88 × 0.95 ÷ 0.9 = 92.9 s).

TEACH IT TO A 14-YEAR-OLD (10 min)

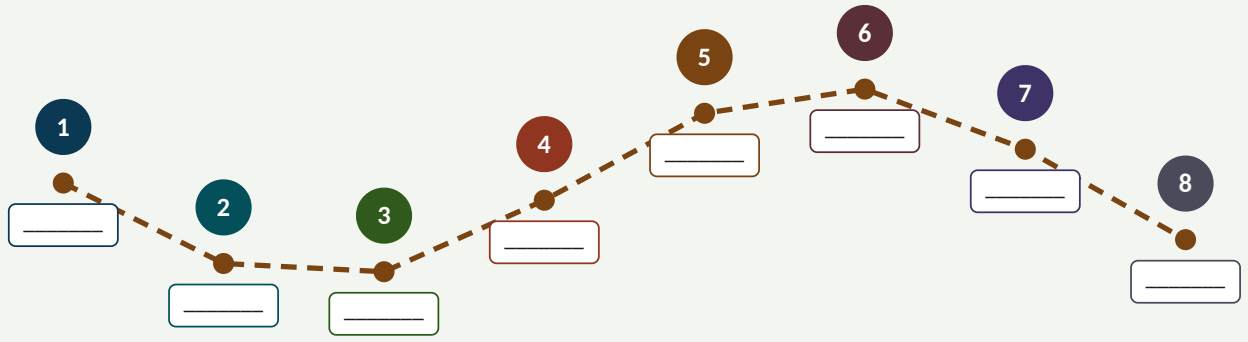
Explain to a 14-year-old why a group of traffic lights shares one rhythm, and why it waits for several agreeing readings before changing its plan.

Words you may not use: saturation, subsystem, hysteresis, offset, tactical

Fill the blank route

Write the concept held at each stop, and its key number.

A walk around a harbour city, from the ferry wharf to the control tower



One-page cheat sheet

1. Subsystems and the critical intersection

Subsystem = 1-10 signals, one cycle. CI = busiest; its DS sets the cycle. DS = (g − (T − nh))/g, h = 1.6 s. Loop each cycle: DS/VK → cycle → splits → offsets → marriage.

Route: Ferry wharf

2. Cycle length: bounded steps towards a target DS

desired = C·DS/0.9. Step +6 (+9 if DS ≥ 1), −6. LMIN 40, LMAX 140, stretch 110. Example: 80 s, DS 1.05 → 89 s; then DS 0.97 → 95 s.

Route: Opera house

3. Split plans: equalise DS, then vote

$DS_{plan} = DS \cdot g_{now}/g_{plan}$. Pick min of max DS. Vote 2 of 3. Example: 0.95/0.50/0.70 at 40/15/20 → SP2 45/12/18 gives max 0.84.

Route: Fish market

4. Offsets and green waves

OP1 = +x/v, OP2 = −x/v (mod C), OP3 = 0, OP4 = C/2 alternate. 300 m at 13.9 m/s = 21.6 s. Heavier if > 1.3× other. Vote 4 of 5.

Route: Harbour bridge

5. Marriage and divorce

Marry: VK ≥ 18 and |ΔC| ≤ 25 s, 3 votes in a row → both use max cycle. Divorce: VK < 10, 3 votes. Hysteresis = different on/off thresholds.

Route: Wedding chapel

6. Tactical control: gap-out, skipping, holding

Gap-out: $t \geq g_{min}$ and gap ≥ 3 s. Unused time passes on. Skip phases with no demand. Coordinated (main) phase held. Example: cars to 5.5 s, g_{min} 8 → ends 8.5 s, 16.5 s saved.

Route: Lighthouse

7. Public transport priority

High: serve ASAP, skip phases (trams, emergency). Medium: extend green or shorten others (buses). Low: normal sequence. Example: arrive 42 s → 38 / 18 / 3 s delay. Lab A1 bus delay: fixed 110.8, actuated 51.8, SCATS 29.7 s.

Route: Tram stop

8. Operating modes and the full system on SUMO

Masterlink (adaptive) → Flexilink (time-of-day + synced clocks) → Isolated (local only) → flashing. Lab A1: 77.2 / 40.9 / 36.9 s; stops 1.82 / 1.57 / 1.30. Ablation off: tactical 62.0, marriage 50.3, offset 45.2, bus 42.5, split 37.5, cycle 34.4.

Route: Control tower

Numbers to remember

- Target DS 0.9; h = 1.6 s
- Cycle steps +6 (+9 if DS ≥ 1), −6; LMIN 40, LMAX 140, stretch 110 s
- Split vote 2 of 3; offset vote 4 of 5
- Marry VK ≥ 18, divorce < 10, 3 votes, cycles within 25 s
- Gap setting 3 s; main road held
- Lab A1 delay 77.2 / 40.9 / 36.9 s; bus 110.8 / 51.8 / 29.7 s
- Ablation off: tactical 62.0, marriage 50.3, offset 45.2 s

Answer key

Try it now

Try it 1 (Subsystems and the critical intersection): n × h = 32 s. Wasted = 18 − 32 = −14 s. Used = 40 + 14 = 54 s. DS = 54 ÷ 40 = **1.35**. DS can exceed 1 when vehicles pass with gaps shorter than h: this junction is very busy.

Try it 2 (Cycle length: bounded steps towards a target DS): Desired = 120 × 0.80 ÷ 0.9 = 106.7 s. Gap −13.3 s, limit −6 s, so C_{next} = **114 s**. Stretch above 110 s = **4 s**, which all goes to the main-road phase.

Try it 3 (Split plans: equalise DS, then vote): A: 0.98 × 30 ÷ 36 = **0.82**. B: 0.60 × 30 ÷ 24 = **0.75**. Highest 0.82 < 0.98, so **P is better** (it would still need to win the vote).

Try it 4 (Offsets and green waves): 0, 21.6, 43.2, 64.7 s; the last wraps: 64.7 − 60 = 4.7 s. So **0, 21.6, 43.2, 4.7 s**.

Try it 5 (Marriage and divorce): **No.** VK is high, but the cycles differ by 30 s, more than the 25 s allowed, so the votes do not count.

Try it 6 (Tactical control: gap-out, skipping, holding): At 5.0 s the gap is 3.0 s, but that is before the 6 s minimum. At 6.0 s the gap is 4 s, so the green would end at **6.0 s**; the car at 6.5 s just misses. (If it had come at 5.5 s, the timer would have reset and the green would end at 8.5 s.)

Try it 7 (Public transport priority): Medium can **extend** the current A green by a few seconds so the bus passes. Bus delay ≈ **0 s**; B starts a little later and is shortened by the same few seconds.

Try it 8 (Operating modes and the full system on SUMO): (1.82 − 1.30) ÷ 1.82 = 0.52 ÷ 1.82 = **29%** fewer stops.

Self-quiz

Q1: B. The CI, usually the busiest. The average would starve the busiest junction. (Concept 1)

Q2: B. 2 of 3 for splits; 4 of 5 is for offsets; 3 in a row is marriage. (Concept 3)

Q3: C. The larger, so the busier subsystem is not starved. (Concept 5)

Q4: A. High priority serves the transit phase as soon as it is safe. (Concept 7)

Q5: B. Flexilink runs time-of-day plans with synced clocks so offsets still hold. (Concept 8)

Q6: B. Desired 133.3 s; DS ≥ 1 allows +9, so 109 s. 106 s uses the normal +6 limit. (Concept 2)

Q7: C. Desired 30 s, but the step down is at most 6 s, so 54 s. 40 s is LMIN, not reached yet. (Concept 2)

Q8: A. 0.90 × 30 ÷ 36 = 0.75. 1.08 multiplies the wrong way round. (Concept 3)

Q9: A. SCATS minimises the highest DS: X's worst is 0.85, Y's is 0.95. Averages are a trap. (Concept 3)

Q10: C. 800 ÷ 16 = 50 s, below 90 so no wrap. 25 s is one link only. (Concept 4)

Q11: C. Neither direction is more than 1.3× the other, so simultaneous or alternate, depending on travel time vs C/2. (Concept 4)

Q12: B. Divorce needs VK below 10; between 10 and 18 nothing changes (hysteresis). (Concept 5)

Q13: B. The gap reached 3 s at 5 s, but the green cannot end before the 8 s minimum. (Concept 6)

Q14: B. Holding it keeps progression; spare time returns to it. (Concept 6)

Q15: B. 'As soon as safely possible': minimum greens, pedestrian clearance and intergreens are still respected. (Concept 7)

Q16: B. Ablation results depend on the corridor and demand; the gain appears when demand moves far from the fixed value. (Concept 8)

Q17: B. 40 s apart exceeds the 25 s compatibility limit in this lab. (Concept 5)

Q18: B. Switching them off raised delay to 62.0 and 50.3 s, the two largest increases. (Concept 8)

Draw it: the completed flow is Figure 1.

Your revision plan

When	What	Done
Today	Read one concept a sitting. Do every Try-it-now on paper.	☐
Day 1	Walk the route from memory. Redraw the master mind map.	☐
Day 3	Warm-up and core quiz without notes. Re-read what you missed.	☐
Day 7	Whole quiz. Do the 'build the smallest example' task.	☐
Day 14	Teach-it task out loud. Cheat sheet from memory, then compare.	☐
Before an exam or project	Cheat sheet plus route walk only.	☐

Going further

- Sims, A.G. and Dobinson, K.W. (1980). The Sydney Coordinated Adaptive Traffic (SCAT) system philosophy and benefits. *IEEE Transactions on Vehicular Technology*, 29(2), 130-137.
- Lowrie, P.R. (1982). The Sydney Coordinated Adaptive Traffic System: principles, methodology, algorithms. *IEE International Conference on Road Traffic Signalling*, London.
- Transport for NSW: SCATS product information, scats.nsw.gov.au.
- This lab's website: Advanced page and SUMO labs A1-A2, smrahmanbd.github.io/scats-learning-lab.

HONESTY NOTE

SCATS is owned by Transport for NSW. This guide describes an independent, educational re-implementation of concepts published in the literature. Parameter values (step limits, voting windows, marriage thresholds, gap setting) are this lab's illustrative teaching values, not official SCATS settings. Worked examples use invented numbers; the bus priority example is illustrative. Lab A1-A2 results come from this repository's SUMO runs (seed 1).